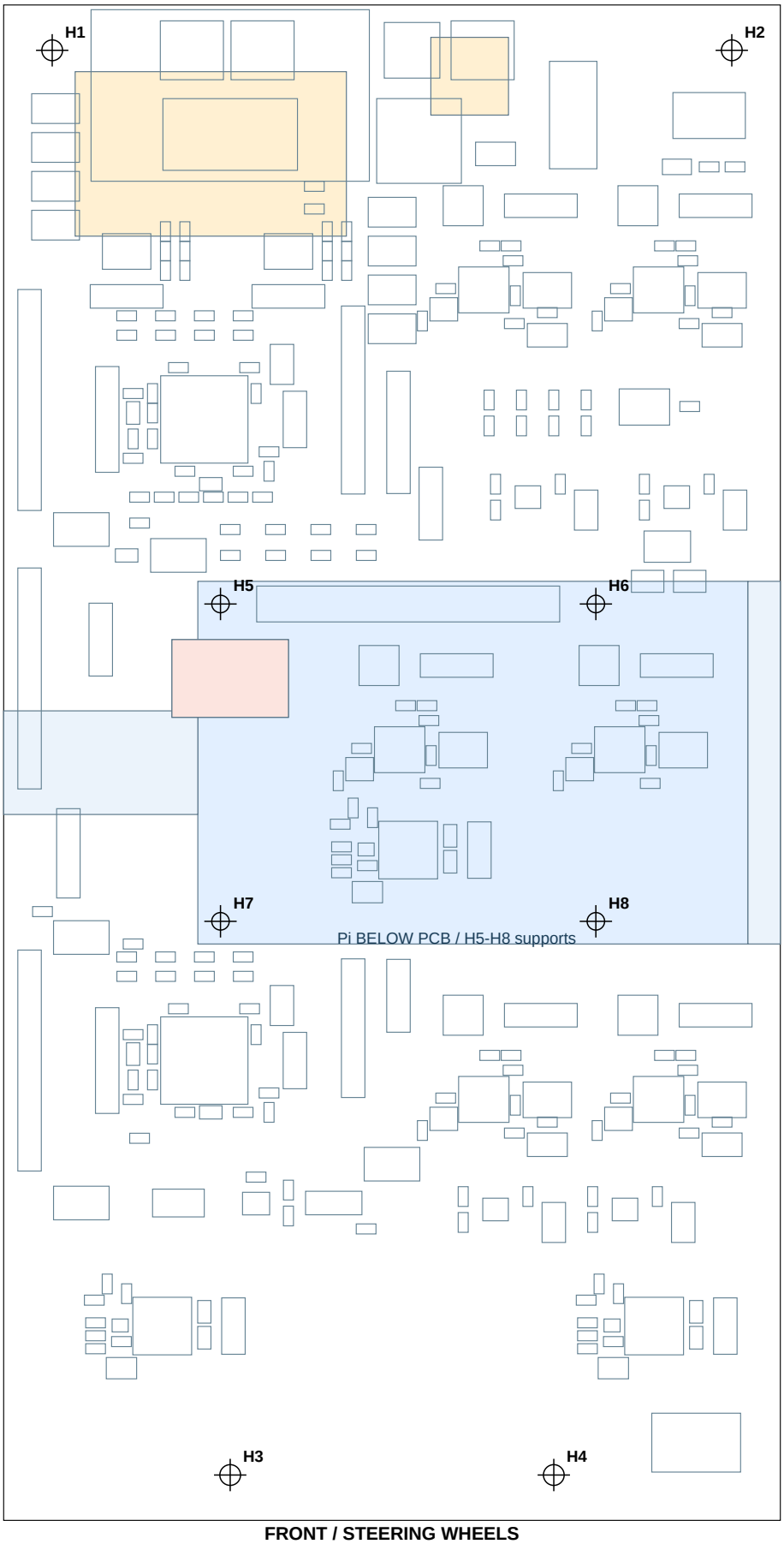
# Lab 4 - provisional PCB fit template

Print at 100% / Actual size. Check the 50 mm bar before comparing with the car.

**MEASURED HOLE CENTERS - PLACEMENT PREVIEW - NOT FOR FABRICATION**

**REAR / DRIVE MOTORS (0,0) at upper-left; x right, y down**




**FRONT / STEERING WHEELS**

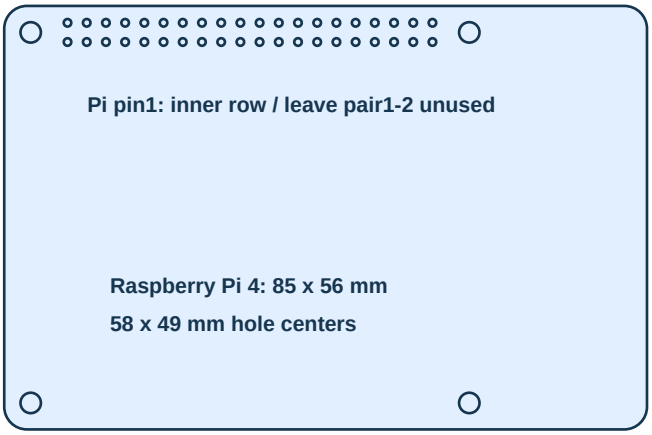
H1-H4: measured centers; M3 PCB drills proposed. Chassis holes measured 4.5 mm.

Check wires, steering travel, Pi below the PCB, port access and separate battery location.

This bar must measure 50 mm.

# Pi alignment and vertical fit check

Top view below is 1:1. Place the unpowered Pi component-side-up to compare holes and pins.




USB / Ethernet face right.

GPIO row faces the REAR.

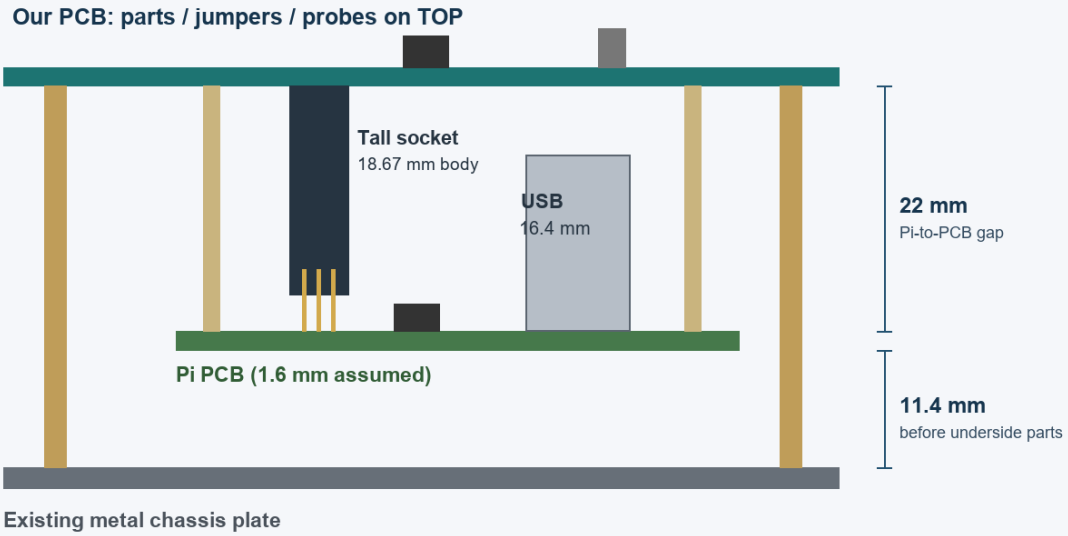
Socket mates Pi pins3-38 only.

Do not use this as a drill guide.

Main-PCB Pi origin: (30,89) mm.

## How the boards stack - proposed dimensions

Pi component side faces UP. J70 socket is soldered to the UNDERSIDE of our PCB.




35 mm chassis standoffs proposed. Caliper heights recorded; final mating is checked at assembly.

### Recorded geometry and remaining fit checks:

Chassis spacers: 35 mm nominal; four-hole paper fit passed with ~50 mm spacers.

Pi-to-PCB gap: 22 mm nominal; calculated socket engagement 5.17 mm.

Reported underside parts <5 mm; calculated nominal metal clearance 6.4 mm.

USB top measured ~18 mm from Pi board bottom; check any added cooler.

Chassis and Pi hole patterns checked; repeat only if the layout changes.

Still check unknown plug/cable/steering envelopes and assembled socket fit.

Pi hardware OD <=5.5 mm. All holes: 3 mm from hole edge. Socket first pair aligns to Pi3/4, not1/2.